Value at Risk (VaR) Forecasting and Diagnostic Validation

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Research question

Does Filtered Historical Simulation improve VaR forecasting across different market regimes, and can we accurately backtest the results?

Financial implications

- Underestimating VaR leads to insufficient capital to cover risks.
- Clustered breaches indicate that the model does not respond quickly enough to market stress.

Expectations

- We expect Filtered Historical Simulation to outperform the other models in backtests.

Literature

Barone-Adesi et al. (1999)

Filtered Historical Simulation · Resample shocks after filtering volatility.

Diebold, Gunther & Tay (1998)

Probability Integral Transform · Evaluate the entire forecast distribution.

Kupiec (1995)

Unconditional coverage · Is the exception frequency correct?

Christoffersen (1998)

Independence and conditional coverage · Are exceptions clustered?

VaR forecasts and backtesting

1 · Forecast tomorrow's loss threshold

- **Variance–covariance:** zero-mean Normal law with rolling volatility.
- **Historical Simulation (HS):** empirical distribution of the last 250 returns.
- **Filtered Historical Simulation (FHS):** resampled shocks with GJR-GARCH volatility.

2 · Record each exception

$$I_t = \mathbf{1}\{r_t < -\mathrm{VaR}_{c,t}\}, \qquad \Pr(I_t = 1) = 1 - c$$

t: forecast day; r: portfolio return; c: confidence (95% or 99%); VaR: positive loss threshold.
I = 1 for a breach, 0 otherwise; the indicator 1condition equals 1 when the condition holds.

Test at the 5% significance level (reject if p < 0.05)

- **Kupiec / unconditional coverage (UC):** correct exception frequency.
- **Christoffersen / independence (IND):** no clustering.
- **Joint conditional coverage (CC):** UC + IND.

Filtered Historical Simulation

Why go beyond the alternatives?

- **Parametric (variance–covariance):** fast and analytical, but Normality misses fat tails.
- **Historical Simulation:** uses real data and equal weights, but reacts slowly to shocks.

FHS: a semiparametric combination

- Filter historical returns with a conditional volatility model, removing volatility clusters.
- Capture leverage with GJR-GARCH: losses can raise volatility more than gains.
- Standardise historical shocks, treat them as iid, then reapply next-day volatility.

Keep empirical tails; adapt their scale to current risk

- Retain historical extreme events without assuming a Normal return distribution.
- Scale past shocks dynamically as market volatility changes.

Filter volatility → **Standardise** → **Resample** → **Rescale**

FHS · GJR-GARCH filtering and simulation

1 · Assume zero conditional mean

$$\mu_t = \mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = 0 \quad \Longrightarrow \quad r_t = \mu_t + \epsilon_t = \epsilon_t$$

Given past information, the expected return is zero. Realised returns still vary.

2 · Filter variance: GJR-GARCH(1,1)

$$\sigma_t^2 = \omega + \beta\sigma_{t-1}^2 + \alpha\epsilon_{t-1}^2 + \gamma\epsilon_{t-1}^2 I_{t-1}$$

σ_t^2 : conditional variance

ϵ_{t-1} : previous return / shock

$I_{t-1} = \mathbf{1}_{\{\epsilon_{t-1}<0\}}$: negative-shock indicator

ω : intercept; α : shock impact

β : volatility persistence

γ : extra impact of negative shocks

$$\omega > 0, \quad \alpha, \beta \geq 0, \quad \alpha + \gamma \geq 0; \qquad \alpha + \beta + \frac{\gamma}{2} < 1$$

Stationarity condition for symmetric standardised innovations.

3 · Standardise, resample and rescale

$$z_t = \frac{r_t}{\sigma_t} = \frac{\epsilon_t}{\sigma_t}$$

$$\widetilde{r}_{t+1} = \widetilde{\epsilon}_{t+1} = \widetilde{\sigma}_{t+1}\,\widetilde{z}$$

Resample a historical standardised error; apply next-day conditional volatility.

Tilde denotes a simulated quantity; the simulated shock is the simulated return.

FHS · Advantages and disadvantages

Advantages

- **Responsiveness:** today's crash can raise tomorrow's volatility estimate and VaR.
- **No Normal return-distribution assumption:** retain the empirical tails.
- **Leverage effect:** losses can create more uncertainty than gains.

Disadvantages

- **Long computation times:** rolling estimation and resampling.
- **Historical dependence:** a quiet sample can produce overly optimistic paths.
- **Extreme stress and outliers** can destabilise parameter estimation.

Why GJR-GARCH? ARCH/GARCH is symmetric; GJR allows a different response to negative shocks.

PIT & Rosenblatt Transformation

Probability Integral Transform

$$U = F_X(X) \sim \text{Uniform}(0, 1)$$

X : random return; F_X : its true continuous CDF

U is the return's percentile in its distribution: values between 0 and 1 are equally likely.

Conditional PIT: the Rosenblatt transformation

Evaluate each realised return using its conditional forecast CDF.

$$u_t = \widehat{F}_{t|t-1}(r_t), \qquad t = 1, \ldots, T$$

r_t : realised portfolio return on day t

$\widehat{F}_{t|t-1}$: forecast CDF for day t, using information through day $t-1$

u_t : percentile of the realised return within its forecast distribution

t : evaluated day; T : total number of evaluated forecasts

Past returns inform each forecast, allowing volatility to change over time.

PIT · Calibration and Normal scores

Under correct conditional forecasts

If the continuous, one-step-ahead conditional forecast distributions are correct:

$$\{u_t\}_{t=1}^{T} \overset{\text{iid}}{\sim} \text{Uniform}(0,1)$$

iid: independent and identically distributed.
Uniform percentiles, with no dependence over time.

Normal scores and Q–Q diagnostics

$$z_t = \Phi^{-1}(u_t) \overset{\text{iid}}{\sim} N(0,1) \qquad \text{under the same assumption}$$

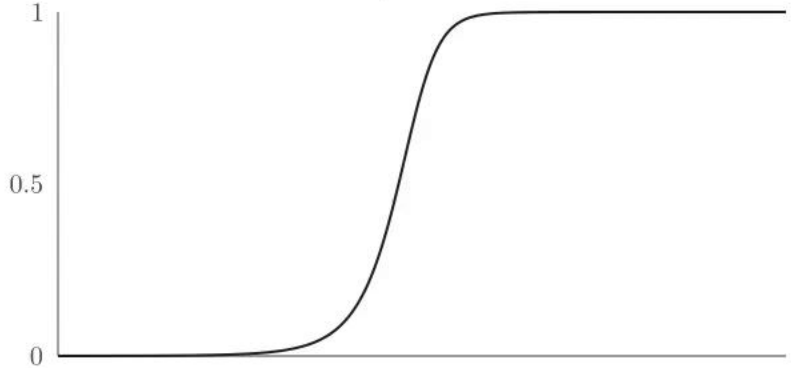
z_t : transformed score for day t
Φ : standard Normal CDF; Φ^{-1} : its inverse, or quantile function
$N(0,1)$: Normal distribution with mean 0 and variance 1

- Compare ordered scores with standard Normal quantiles; departures reveal calibration errors.
- Uniformity concerns the distribution.
- Independence concerns patterns over time: test it separately.

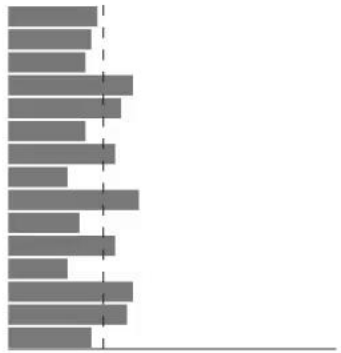
PIT · Changing predictive distributions

t: day; r: realised return; u: forecast percentile; z: Normal score; Phi: standard Normal CDF.

$\widehat{F}_{t|t-1}(r)$

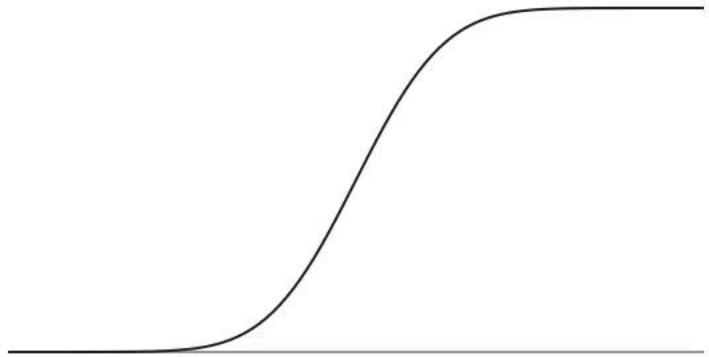


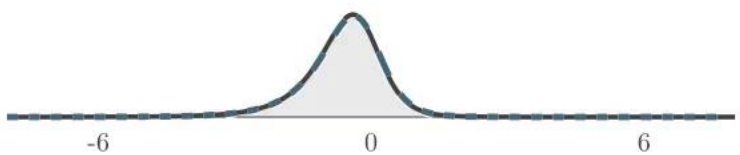
$u_t = \widehat{F}_{t|t-1}(r_t)$



$U(0,1)$

$z_t = \Phi^{-1}(u_t)$



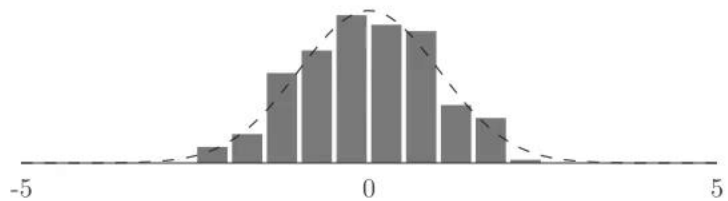


Return densities

Simulated returns

True distribution (unobserved)

Predicted distribution



Normal scores $N(0,1)$

$t = 1$

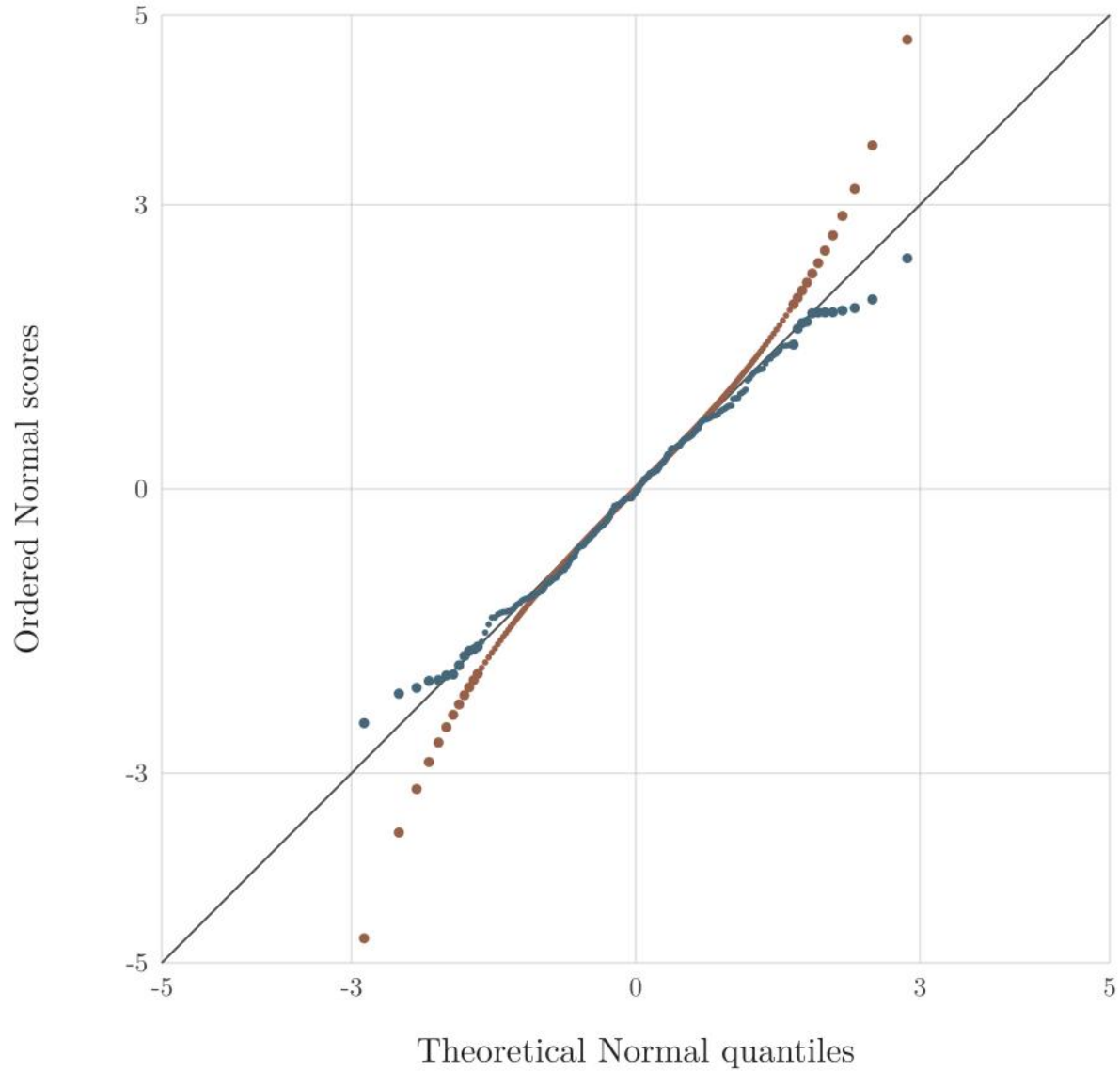
$t = T$

Normal Q–Q and tail misspecification

Approximate Normal calibration

Tail misspecification

Heavier losses



Heavier gains

Two evaluation windows · 11 euro-area equities

Stable: Jan 2014–Dec 2017 · 1,025 days Stress: Jan 2018–Dec 2021 · 1,026 days

Fixed portfolio weights · 250 past trading days per forecast · one-day horizon

No.	Company	Weight
01	Deutsche Telekom	18.3%
02	Siemens	15.2%
03	Banco Santander	13.8%
04	Sanofi	13.8%
05	ASML	8.0%
06	Allianz	6.1%
07	AB InBev	5.6%
08	TotalEnergies	5.5%
09	L'Oréal	5.2%
10	Iberdrola	4.4%
11	Deutsche Börse	4.1%
		100%

* Returns: percentage change in close prices. Source: Yahoo Finance. Weights sum to 100%.

Backtesting: 2014–2017

1,025 daily forecasts · p-values · UC: frequency · IND: independence · CC: joint coverage

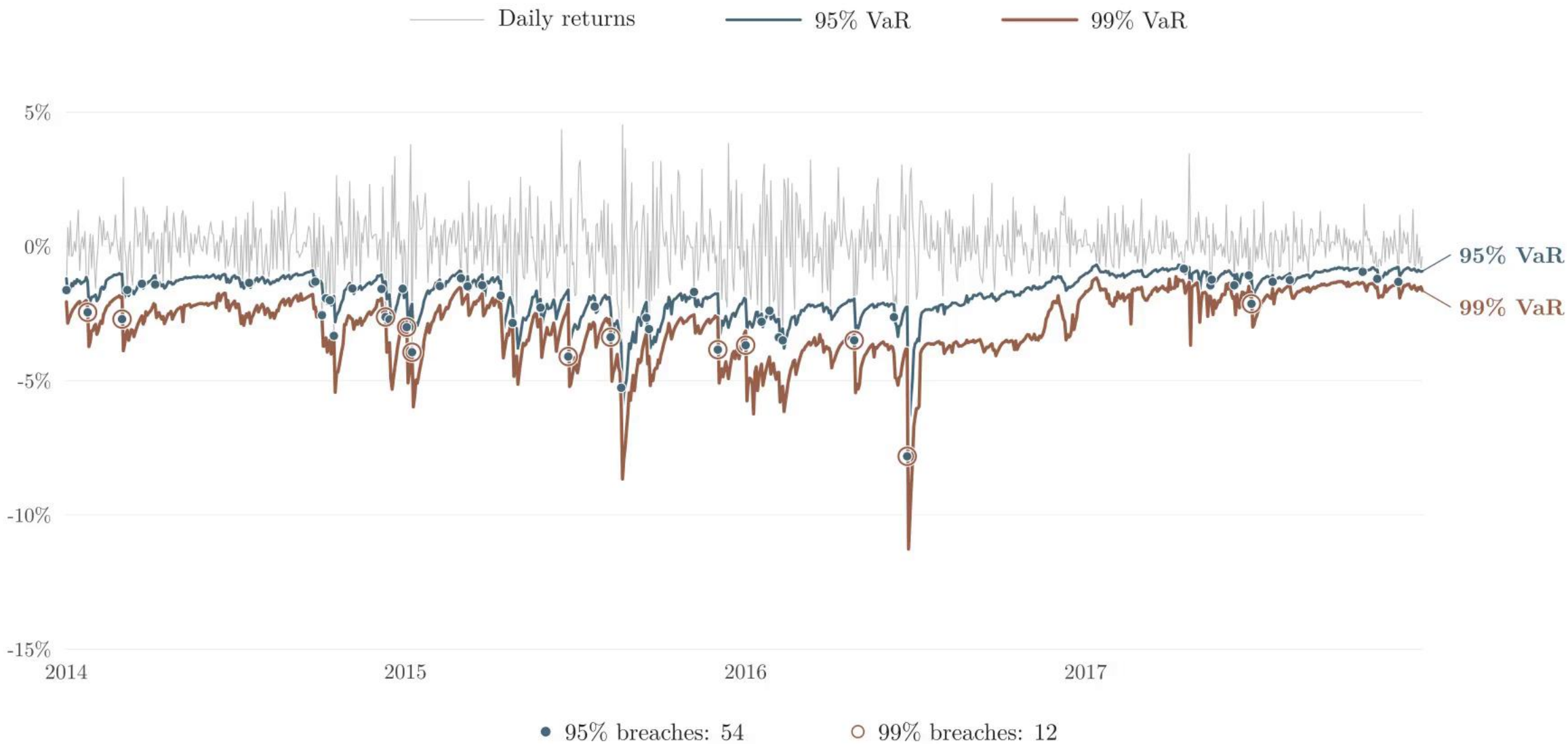
95% VaR	Model	UC	IND	CC
	Normal	0.444	0.194	0.321
	HS	0.595	0.233	0.426
	FHS-GJR	0.696	0.197	0.403

99% VaR	Model	UC	IND	CC
	Normal	0.053	0.449	0.115
	HS	0.407	0.563	0.600
	FHS-GJR	0.593	0.594	0.752

No model is rejected by these tests.

p: probability of equally or more extreme evidence under the null; underlined $p < 0.05 \Rightarrow$ reject.

FHS-GJR · 2014–2017



Backtesting: 2018–2021

1,026 daily forecasts · p-values · UC: frequency · IND: independence · CC: joint coverage

95% VaR	Model	UC	IND	CC
	Normal	0.347	**<.001**	**<.001**
	HS	0.422	**<.001**	**0.003**
	FHS-GJR	0.137	0.892	0.328

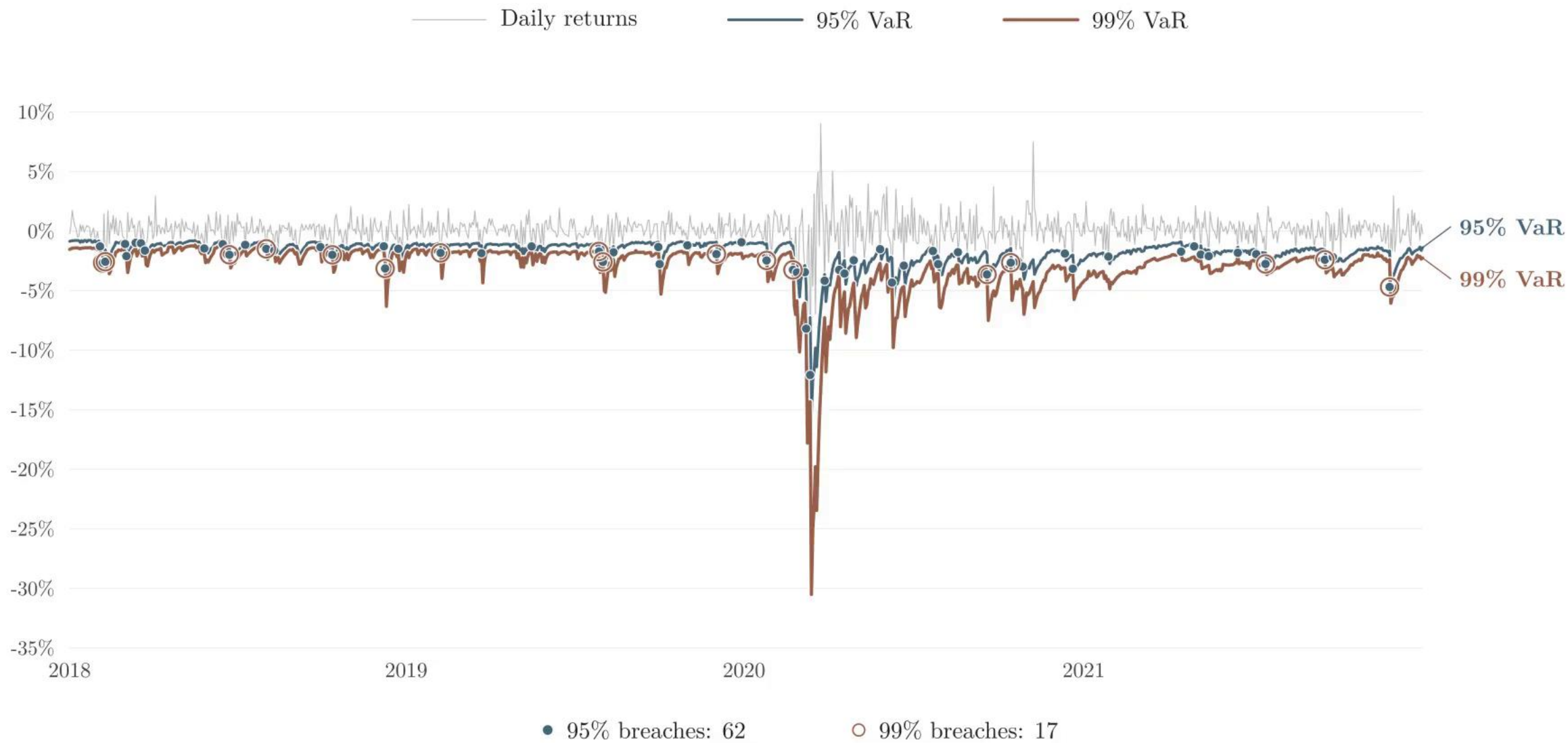
99% VaR	Model	UC	IND	CC
	Normal	**<.001**	**0.003**	**<.001**
	HS	0.164	**0.017**	**0.022**
	FHS-GJR	0.053	0.449	0.116

FHS-GJR alone avoids rejection in every test at both levels.

p: probability of equally or more extreme evidence under the null; underlined $p < 0.05 \Rightarrow$ reject.

99% GJR Kupiec: $p = 0.0533$ · 17 breaches vs 10.26 expected — close to rejection.

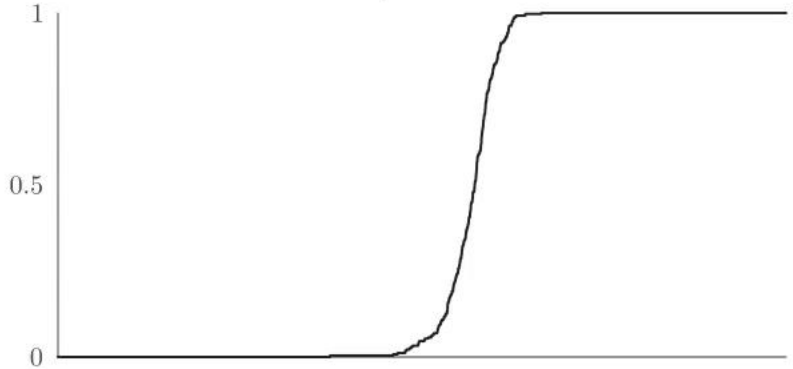
FHS-GJR · 2018–2021



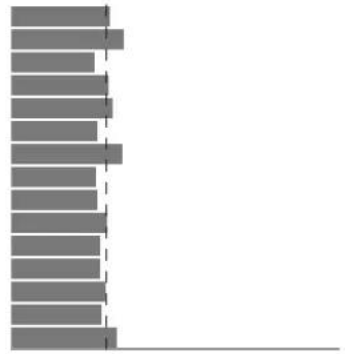
FHS-GJR · Crisis PIT · 2018–2021

t: day; r: realised return; u: forecast percentile; z: Normal score; Phi: standard Normal CDF.

$\widehat{F}_{t|t-1}(r)$

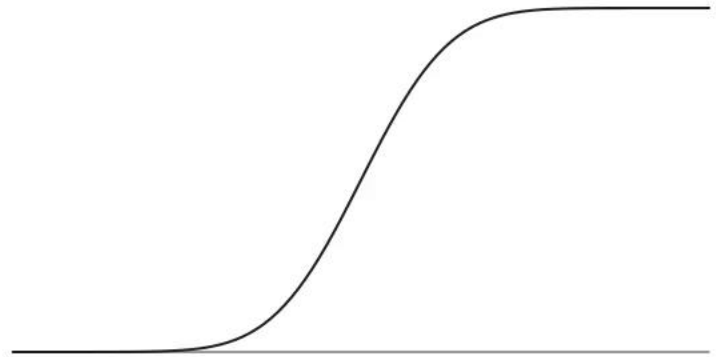


$u_t = \widehat{F}_{t|t-1}(r_t)$

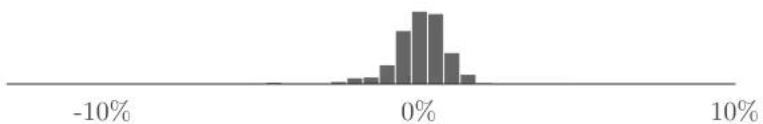


$U(0,1)$

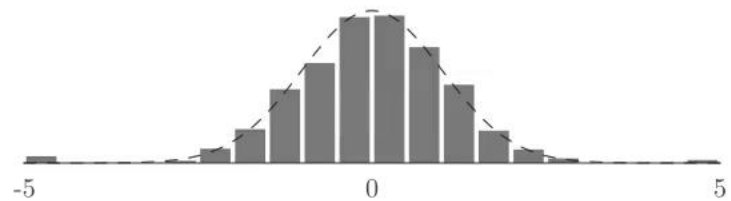
$z_t = \Phi^{-1}(u_t)$



Each dot uses its own dated forecast.



FHS-GJR · Forecast density



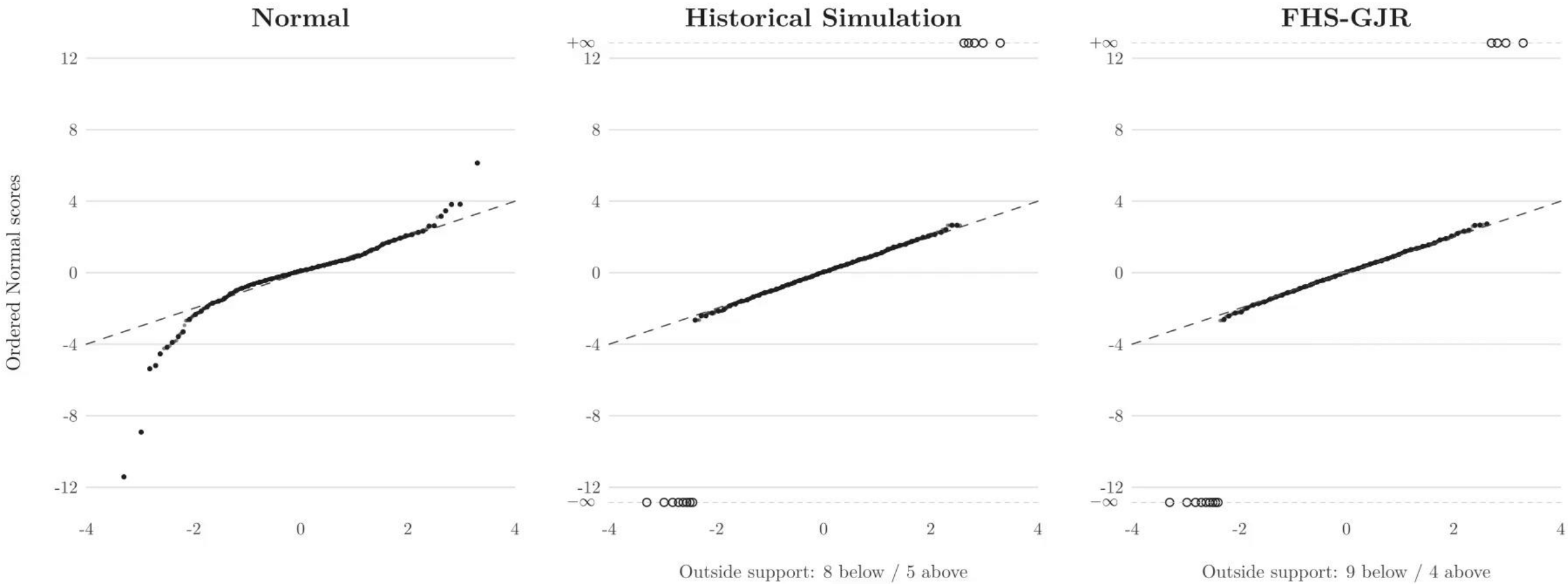
Normal scores $N(0,1)$

2021-12-31



Q–Q comparison · Crisis 2018–2021

1,026 observations



Findings

FHS-GJR improves VaR forecasting under stress: alone not rejected in all tests at 95% and 99% (2018–2021).
In calm markets, tests do not discriminate (2014–2017); inference is specific to these data and windows.

01 · Stable market

- No model is rejected: the tests do not discriminate among forecasts.

02 · Stress market

- Only asymmetric FHS-GJR avoids rejection at both 95% and 99%.
- Leverage makes negative shocks raise forecast volatility more strongly.

03 · How convincing is the evidence?

- 99% GJR Kupiec p = 0.0533: 17 breaches against 10.26 expected.
- Borderline non-rejection supports caution, not definitive validation.

Research limitations

01 · Information arrives after the shock

Past returns only; no option-implied volatility to anticipate a regime change.

02 · Two calendar windows

One stable and one stress period; the result need not generalise to other episodes.

03 · Concentrated portfolio and limited model set

11 stocks, fixed weights, possible selection and survivorship bias.

Filtering is at portfolio level; time-varying asset correlations are not modelled.

04 · One-day validation

Rare 99% breaches limit statistical power. Pooled Q–Q plots do not test independence.

Expected Shortfall (ES), the mean loss beyond VaR, is not backtested.

Thank you

Animation & presentation

Manim

github.com/ManimCommunity/manim

Manim Slides

github.com/jeertmans/manim-slides

Presentation extension for Manim

Code & research

Project repository

github.com/mattiakeppel/VaR-methods



Repository: code, data and slides

Presentation coded using Codex